

YAUVI Structural Workbench: evidence-bounded, reproducible structural protein analysis

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Software

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Summary

YAUVI Structural Workbench is a local Python software suite and browser interface for six common structural-protein questions: coordinate identity and quality, membrane orientation, conformational resemblance, functional-site context, biological assembly interfaces, and comparative structural/sequence relationships. Each analysis remains an independently installable command-line package. The workbench creates typed configurations, verifies files by SHA-256, invokes the registered command, and renders its JSON and tabular evidence into a readable, printable report.

The central design constraint is that evidence dimensions are not collapsed into a universal protein or function score. Unknown provenance remains unknown; active-like describes coordinate resemblance rather than biochemical activity; an annotated catalytic site is not observed catalysis; and a shared fold is not exact functional transfer. Missing references, mappings, runtimes, or validation reports remain visible as missing or scientifically incomplete evidence.

Statement of need

Structural bioinformatics typically requires moving between coordinate validation, residue numbering systems, sequence annotations, reference conformations, biological assemblies, surface areas, and similarity searches. Author and label residue identities in PDBx/mmCIF are particularly important when a measurement must trace back to an exact sequence position. Community validation systems such as wwPDB and MolProbity provide essential model-quality evidence ([Williams & others, 2018](#); [Worldwide Protein Data Bank, n.d.](#)), while Gemmi provides robust macromolecular format support ([Wojdyr, 2022](#)). However, these results are often reviewed in separate tools, and missing evidence can be mistaken for a favorable or negative result.

YAUVI provides one reproducible case boundary while leaving scientific calculations in standalone packages. It is intended for structural biologists, bioinformaticians, and trainees who need inspectable evidence and explicit claim limits rather than an opaque ranking. The same deterministic reports can support method development, target characterization, comparison of experimental and predicted structures, and planning of follow-up validation without generating a wet-lab protocol.

State of the field and build-versus-contribute justification

Existing programs already provide stronger specialist calculations than a new implementation should attempt to replace. Gemmi supplies standards-aware coordinate parsing ([Wojdyr, 2022](#)); wwPDB and MolProbity supply community model validation ([Williams & others, 2018](#);

39 [Worldwide Protein Data Bank, n.d.](#)); OPM/PPM supplies a richer membrane-positioning
40 reference system ([Lomize et al., 2006](#)); and Foldseek and DIAMOND supply established
41 structure and sequence searches ([Buchfink et al., 2021](#); [Kempen & others, 2024](#)). YAUVI
42 therefore invokes or imports these methods through named adapters and preserves their scores,
43 parameters, units, versions, and limitations rather than reimplementing them.

44 The distinct contribution is the evidence boundary joining these methods: exact input identity,
45 residue mapping, explicit provenance, fail-closed preflight, separation of scientific dimensions,
46 deterministic reports, and claim ceilings shared across six independently runnable workflows.
47 Extending a single existing specialist package would not provide this cross-method case and
48 reporting contract, while replacing the specialist algorithms would reduce scientific quality and
49 maintenance sustainability.

50 Software design

51 The workbench stores input bytes content-addressably and creates immutable run records.
52 Task definitions declare the scientific question, accepted artifact types, format validator, source
53 assistance, missing-evidence consequence, outputs, and claim ceiling. A source finder links
54 input roles to official RCSB PDB, wwPDB validation, AlphaFold DB, UniProt, SIFTS, M-CSA,
55 PDB CCD, ChEBI, and OPM/PPM resources. Network access is disabled by default. When
56 explicitly enabled, only registered artifact types and public identifiers can be acquired; cache
57 acquisition and adoption into an analysis are separate operations.

58 StructQC establishes coordinate provenance and residue identity before composed workflows.
59 MembraneOrient estimates a context-declared membrane frame, with OPM/PPM retained as an
60 external comparison standard ([Lomize et al., 2006](#)). StateAtlas uses sequence-mapped Kabsch
61 alignment, RMSD/RMSF, deterministic clustering, and two-sided experimental references.
62 SiteContext and ActState keep annotation, observed chemistry, and geometric competence
63 separate. AssemblyContext reports heavy-atom contacts, stoichiometry evidence, and method-
64 specific solvent accessible surface area. SF-CSA executes Foldseek ([Kempen & others, 2024](#))
65 and DIAMOND ([Buchfink et al., 2021](#)) as separate structural and sequence legs against
66 checksum-pinned reference universes.

67 Completed or scientifically incomplete runs emit `REPORT_DATA.json`, printable `REPORT.html`, a
68 deterministic `RAW_EVIDENCE.zip`, checksums, and the canonical run manifest. Display rounding
69 never modifies the underlying scientific values.

70 Validation and research use

71 The suite uses synthetic offline fixtures, schema tests, fail-closed boundary tests, transformation
72 and ordering invariance tests, deterministic output comparisons, controller security tests, and
73 standalone package tests. External scientific qualification is reported separately from software
74 correctness. The first checksum-locked public collection includes wwPDB validation, OPM
75 membrane strata, KinCore-labeled two-sided conformational references, an M-CSA enzyme case,
76 a deposited assembly evaluated with FreeSASA, and a CATH-labeled SF-CSA mini-database
77 searched by Foldseek and DIAMOND. Four public cases pass their predeclared gates and two
78 remain partial. No release qualification is claimed until all six workflow gates pass and a
79 broader stratified evaluation supports generalization beyond the named cases.

80 Research impact statement

81 At this pre-public stage, the project has not recorded independent adoption, published research
82 use, or six completed external benchmark gates. The software is therefore not presented as
83 JOSS submission-eligible. Current reproducible benchmark records expose both successful cases

84 and scientific limitations; they do not substitute for documented use in real structural-biology
85 analyses, independent installation feedback, and public issue-driven refinement. Aspirational
86 utility is not counted as realized research impact in the release state.

87 Limitations

88 The workbench does not replace experimental validation, density inspection, biochemical
89 assays, native-surface measurements, or expert curation of conformational references. Optional
90 external executables and databases retain their own licenses and must be acquired independently.
91 Predicted models cannot establish activity, assemblies may be context-dependent, solvent-
92 accessible area is method-specific, and sequence or structure similarity alone cannot transfer
93 mechanism.

94 AI usage disclosure

95 OpenAI Codex using GPT-5-family coding models assisted with code generation, refactoring,
96 interface copy, documentation, test scaffolding, and manuscript drafting during private
97 development. Anthropic Claude also assisted during earlier private development, but its
98 exact model/version record has not yet been recovered; this is a release blocker that must be
99 resolved before submission. The human author reviewed and edited the assisted outputs, ran
100 the recorded software checks, selected the scientific boundaries and sources, and made the core
101 design decisions. The human author remains responsible for originality, accuracy, licensing,
102 ethical and legal compliance, and every manuscript claim. AI-generated suggestions are never
103 treated as scientific evidence or benchmark results. AI tools will not be used for author-editor
104 or author-reviewer conversations except where a journal policy explicitly permits translation.

105 Conflicts of interest and funding

106 Conflict-of-interest and funding statements have not yet been finalized for submission. They
107 must be supplied and approved by every listed author before the paper can leave pre-public
108 preparation.

109 Acknowledgements

110 The project depends on the maintainers and data curators of its open scientific dependencies
111 and public reference resources. Those projects must be cited independently when their methods
112 or data are used.

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